

1 CB2 8.3

$Y_1, \dots, Y_m \sim \text{Bern}(\theta)$. Show that the LRT of $H_0 : \theta \leq \theta_0, H_1 : \theta > \theta_0$ will reject if $\sum_{i=1}^m Y_i > b$

$$L(Y|\theta) = \prod \theta^y (1-\theta)^{1-y} = \theta^{\sum y} (1-\theta)^{m-\sum y}$$

$$\text{LRT} = \frac{\argmax L(\theta|Y)_{\Theta_0}}{\argmax L(\theta|Y)_{\Theta}}$$

The plot of the likelihood is basically a hill, and under the null, the argmax of the likelihood occurs at either θ_0 , or the MLE ($\bar{Y}$) (it is monotonic increasing up to that point). Over the full space, the argmax is at the MLE, $\bar{X} = \frac{\sum Y_i}{m}$

$$\Rightarrow \text{LRT} = \frac{\theta_0^{\sum y} (1-\theta_0)^{m-\sum y}}{(\frac{\sum Y_i}{m})^{\sum y} (1-\frac{\sum Y_i}{m})^{m-\sum y}} = (\frac{m\theta_0}{\sum y})^{\sum y} (\frac{1-\theta_0}{1-\sum y/m})^{m-\sum y}, \text{ we will reject if that's less than } c.$$

$$\ln(\lambda(x)) < \ln(c) \Rightarrow \sum y(\ln(m) + \ln(\theta_0) - \ln(\sum y)) + (m - \sum y)(\ln(1 - \theta_0) - \ln(1 - \sum y/m)) < \ln(c)$$

Let $x = \sum y$:

$$\frac{d}{dx} = \ln(\theta_0) - \ln(1 - \theta_0) - \ln(\frac{x}{m}) - 1 + \ln(\frac{m-x}{m}) + 1 = \ln(\frac{\theta_0(\frac{m-x}{m})}{y/m(1-\theta_0)})$$

In all cases, the fraction inside the log is less than 1, so the ln is negative, meaning this is a decreasing function, such that $\lambda(\sum y) < c \iff \sum Y > b$

2 CB2 8.6

$$X \stackrel{\text{iid}}{\sim} \text{Exp}(\theta); Y \stackrel{\text{iid}}{\sim} \text{Exp}(\mu)$$

a) Find the LRT of $H_0 : \theta = \mu, H_1 : \theta \neq \mu$

$$\text{LRT} = \frac{L(\hat{\theta}_{\Theta_0}|x,y)}{L(\hat{\theta}_{\Theta}|x,y)} = \frac{\sup_{\theta} \frac{1}{\theta^n} e^{-\sum x/\theta} \frac{1}{\mu^m} e^{-\sum y/\mu}}{\sup_{\theta, \mu} \frac{1}{\theta^n} e^{-\sum x/\theta} \frac{1}{\mu^m} e^{-\sum y/\mu}}$$

In the numerator we are assuming that $\theta = \mu$

$$\Rightarrow \frac{\sup_{\theta} \frac{1}{\theta^{n+m}} e^{-(\sum x + \sum y)/\theta}}{\sup_{\theta, \mu} \frac{1}{\theta^n} e^{-\sum x/\theta} \frac{1}{\mu^m} e^{-\sum y/\mu}}$$

$$\text{The numerator: } \frac{d}{d\theta} = \frac{-(n+m)}{\theta^{n+m+1}} e^{-(\sum x + \sum y)/\theta} - \frac{\sum x + \sum y}{\theta^{n+m}} e^{-(\sum x + \sum y)/\theta} = 0 \Rightarrow \hat{\theta}_0 = \frac{\sum x + \sum y}{n+m}$$

In the denominator, the MLEs are just $\bar{x}$ and $\bar{y}$

$$\Rightarrow \frac{L(\hat{\theta}_{\Theta_0}|x,y)}{L(\hat{\theta}_{\Theta}|x,y)} = \frac{(\frac{n+m}{\sum x + \sum y})^{n+m} e^{-\frac{n+m}{\sum x + \sum y}(\sum x + \sum y)}}{(\frac{n}{\sum x})^n e^{-\frac{n}{\sum x} \sum x} (\frac{m}{\sum y})^m e^{-\frac{m}{\sum y} \sum y}} = \frac{(n+m)^{n+m} (\sum x)^n (\sum y)^m}{n^n m^m (\sum x + \sum y)^{n+m}}$$

Thus, LRT says we reject when $\frac{(n+m)^{n+m} (\sum x)^n (\sum y)^m}{n^n m^m (\sum x + \sum y)^{n+m}} \leq c$

b) Show the test in part (a) can be based on statistic $T = \frac{\sum X_i}{\sum X_i + \sum Y_i}$

$$\text{If } T = \frac{\sum X_i}{\sum X_i + \sum Y_i}, \text{ then } 1 - T = \frac{\sum Y_i}{\sum X_i + \sum Y_i}$$

$$\text{We can rewrite that test as: } \frac{(n+m)^{n+m}}{n^n m^m} \left(\frac{\sum x}{\sum x + \sum y} \right)^n \left(\frac{\sum y}{\sum x + \sum y} \right)^m = \frac{(n+m)^{n+m}}{n^n m^m} T^n (1-T)^m$$

c) Find distribution of T when H_0 is true

The sum of exponentials is a Gamma, so in this case $\sum x \sim \text{Gamma}(n, \theta)$; $\sum y \sim \text{Gamma}(m, \mu)$. However, we are assuming the null hypothesis is true, so $\sum y \sim \text{Gamma}(m, \mu)$.

I had to look up what the distribution of $\frac{\text{Gamma}(n, \theta)}{\text{Gamma}(n, \theta) + \text{Gamma}(m, \theta)}$ is, turns out it's $\text{Beta}(n, m)$

Thus, the resulting distribution of T is $\text{Beta}(n, m)$

3 CB2 8.7a

a)

$f(x|\theta, \lambda) = \frac{1}{\lambda} e^{-(x-\theta)/\lambda} I_{[\theta, \infty)}(x)$, both θ, λ unknown. $H_0 : \theta \leq 0, H_1 : \theta > 0$. Find the LRT.

$$L(\lambda, \theta|x) = \prod \frac{1}{\lambda} e^{-(x-\theta)/\lambda} I_{[\theta, \infty)}(x) = \frac{1}{\lambda^n} e^{-(\sum x - n\theta)/\lambda} I_{[\theta, \infty)}(X_{(1)})$$

The marginal effect of θ is that as θ increases (irrespective of λ), the function is increasing (under condition that $\theta < x_{(1)}$). Thus, the $\hat{\theta} = x_{(1)}$

$$\text{For } \lambda: \frac{d}{d\lambda} = \frac{-n}{\lambda} e^{-(\sum x - n\theta)/\lambda} - \frac{\sum x - n\theta}{\lambda^2} e^{-(\sum x - n\theta)/\lambda} = 0 \implies \frac{-n}{\lambda} = \frac{\sum x - n\theta}{\lambda^2} \implies n\lambda = \sum x - n\theta \implies \hat{\lambda} = \bar{x} - \hat{\theta} = \bar{x} - x_{(1)}$$

Those values apply for the denominator. Now for the numerator, we are restricted to $\theta \leq 0$. Thus, if $x_{(1)} > 0$, that is not allowed so the MLE would be 0. So we have $\hat{\theta}_R = \begin{cases} x_{(1)}, & x_{(1)} \leq 0 \\ 0, & x_{(1)} > 0 \end{cases}$

Looking at our above value for λ , we plug in $\hat{\theta}_R = 0$ and get $\hat{\lambda}_R = \bar{x}$

$$\text{Now we do the ratio; } \lambda(x) = \frac{L(\hat{\theta}_R, \hat{\lambda}_R|x)}{L(\hat{\theta}, \hat{\lambda}|x)} = \frac{\frac{1}{\bar{x}^n} e^{-(\sum x - n x_{(1)})/\bar{x}}}{\frac{1}{(\bar{x} - x_{(1)})^n} e^{-(\sum x - n x_{(1)})/(\bar{x} - x_{(1)})}} = \left(\frac{\bar{x} - x_{(1)}}{\bar{x}}\right)^n = \left(1 - \frac{x_{(1)}}{\bar{x}}\right)^n$$

Thus, LRT says we reject when $\lambda(x) \leq c$, which is equivalent to rejecting when $\frac{x_{(1)}}{\bar{x}} \leq a$ for some a .

4 CB2 8.8a or CB2 8.8b

a) $X \sim N(\theta, \alpha\theta)$, θ unknown. Find the LRT of $H_0 : \alpha = 1, H_1 : \alpha \neq 1$

$$L(\theta, a|x) = (2\pi a\theta)^{-n/2} e^{-\frac{1}{2a\theta} \sum (x_i - \theta)^2}$$

$$\implies \ln(L) = -\frac{1}{2} \sum [\ln(2\pi a\theta) + \frac{1}{a\theta} (x_i - \theta)^2]$$

$$\implies \frac{\partial}{\partial a} = -\frac{1}{2} \sum \left[\frac{1}{a} - \frac{1}{a^2\theta} (x_i - \theta)^2 \right] = -\frac{n}{2a} + \frac{1}{2a^2\theta} \sum (x_i - \theta)^2 = 0 \implies \frac{n}{2a} = \frac{1}{2a^2\theta} \sum (x_i - \theta)^2 \implies a = \frac{1}{n\theta} \sum (x_i - \theta)^2$$

$$\implies \frac{\partial}{\partial \theta} = -\frac{n}{2\theta} + \sum \left[\frac{1}{2a\theta^2} (x_i - \theta)^2 + \frac{2(x_i - \theta)}{2a\theta} \right] = 0 \implies \frac{n}{2\theta} = \frac{1}{2a\theta^2} \sum (x_i - \theta)^2 + \frac{n\bar{x} - n\theta}{a\theta}$$

$$\text{Plug the first equation into the second: } \frac{n}{2\theta} = \frac{n\theta \sum (x_i - \theta)^2}{2\theta^2 \sum (x_i - \theta)^2} + \frac{(n\bar{x} - n\theta)n\theta}{\theta \sum (x_i - \theta)^2} = \frac{n}{2\theta} + \frac{(n\bar{x} - n\theta)n}{\sum (x_i - \theta)^2} \implies 0 = \frac{(n\bar{x} - n\theta)n}{\sum (x_i - \theta)^2} \implies \hat{\theta} = \bar{x}$$

$$\text{Plug that back into the first: } a = \frac{1}{n\bar{x}} \sum (x_i - \bar{x})^2 = \frac{\sigma^2}{\bar{x}}$$

Those are the unrestricted MLEs. For the supremums under the null, we let $a=1$. Going back to our previous equations:

$$a = 1 = \frac{n}{\theta} \sum (x_i - \theta)^2 \implies n\theta = \sum (x_i - \bar{x})^2 + n(\bar{x} - \theta)^2 \implies \theta = \sigma^2 + \bar{x}^2 - 2\bar{x}\theta + \theta^2 \implies \hat{\theta}_R = \frac{-1 + \sqrt{1 + 4(\bar{x}^2 + \sigma^2)}}{2}$$

$$\text{This gives the LRT: } \lambda(x) = \frac{L(\hat{\theta}_R|x)}{L(\hat{\theta}, \hat{\lambda}|x)} = \frac{(2\pi\hat{\theta}_R)^{-n/2} e^{-\frac{1}{2\hat{\theta}_R} \sum (x_i - \hat{\theta}_R)^2}}{(2\pi\hat{\theta})^{-n/2} e^{-\frac{1}{2\hat{\theta}} \sum (x_i - \hat{\theta})^2}} = \frac{(2\pi\hat{\theta}_R)^{-n/2} e^{-\frac{1}{2\hat{\theta}_R} \sum (x_i - \hat{\theta}_R)^2}}{(2\pi\hat{\theta})^{-n/2} e^{-\frac{1}{2\hat{\theta}} \sum (x_i - \hat{\theta})^2}} = \left(\frac{\hat{\theta}}{\hat{\theta}_R}\right)^{n/2} e^{(n/2) - \sum \frac{(x_i - \hat{\theta}_R)^2}{2\hat{\theta}_R}} \text{ with } \hat{\theta}_R$$

as described above.

5

$X_i \stackrel{\text{iid}}{\sim} \text{Pois}(\theta), \theta > 0$

a)

Show that LRT for $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$ can be written as $\phi(x) = \begin{cases} 1, & \sum X_i \notin [k_1, k_2] \\ 0, & \sum X_i \in [k_1, k_2] \end{cases}, k_1, k_2 > 0$

$$L(\theta|x) = \prod \frac{\theta^x e^{-\theta}}{x!} = \frac{\theta^{\sum x} e^{-n\theta}}{\prod x!}$$

The unrestricted MLE for $\theta = \hat{\theta} = \bar{X}$

Under the restriction that $\theta = \theta_0$, then $\hat{\theta}_R = \theta_0$

$$\text{LRT} = \frac{L(\hat{\theta}_R|x)}{L(\hat{\theta}|x)} = \frac{\theta_0^{\sum x} e^{-n\theta_0}}{\bar{x}^{\sum x} e^{-n\bar{x}}} = \left(\frac{\theta_0}{\bar{x}}\right)^{\sum x} e^{-n(\theta_0 + \bar{x})}$$

The LRT says we will reject when $\left(\frac{\theta_0}{\bar{x}}\right)^{\sum x} e^{-n(\theta_0 + \bar{x})} \geq a$ or $\leq b$. In this case, this is just an increasing function of $\sum x$, meaning this is equivalent to rejecting when $\sum x \leq k_1$ or $\sum x \geq k_2$.

This can be written as: $\phi(x) = \begin{cases} 1, & \sum X_i \notin [k_1, k_2] \\ 0, & \sum X_i \in [k_1, k_2] \end{cases}$

b)

$n = 5, \alpha = 0.05, \theta_0 = 2$, find k_1, k_2 such that LRT has as close as possible, but not exceeding $\alpha = 0.05$

The sum of Poissons is a Poisson with the sum of the variables, so $\sum x \sim \text{Pois}(\theta_0 * n) = \text{Pois}(10)$

We basically need to solve $P(k_1 \leq \sum x \leq k_2) = 1 - \alpha \implies P(\sum x \leq k_2) - P(\sum x \leq k_1) = 1 - \alpha \implies P(\sum x \leq k_2) - P(\sum x \leq k_1) - 1 + \alpha = 0$.

We can then use root finding to get that value. I used the following r code:

```
func <- function(params) {
  k1 <- params[[1]]
  k2 <- params[[2]]
  ppois(k1, 10) - ppois(k2, 10) - 1 + .05
}
```

```
optim(par = c(1, 20), func)
```

and got a value of: $k_1 = -31.5, k_2 = 47.125$

6

$X = (X_1, X_2, X_3)$ is trinomial: $p = (p_1, p_2, p_3)$ with PMF $f(x|p) = \prod_{j=1}^3 p_j^{x_j}; \sum_{j=1}^3 x_j = m; p_j > 0, \sum p_j = 1$.

Let $m = 1$, and for $\theta \in (-1, 1) : p_1 = \frac{(1-\theta)^2}{4}; p_2 = \frac{(1-\theta^2)}{2}, p_3 = \frac{(1+\theta)^2}{4}$

a)

Let $Y_j = \sum_{i=1}^n X_{ij}$. Show that MLE $\hat{\theta} = \frac{Y_3 - Y_1}{n}$

$$f(x|\theta) = \left(\frac{(1-\theta)^2}{4}\right)^{x_1} \left(\frac{1-\theta^2}{2}\right)^{x_2} \left(\frac{(1+\theta)^2}{4}\right)^{x_3} = \frac{(1-\theta)^{2x_1} (1-\theta^2)^{x_2} (1+\theta)^{2x_3}}{2^{2x_1} 2^{x_2} 2^{2x_3}}$$

$$L(\theta|x) = \prod_{i=1}^n f(x_i|\theta) = \frac{(1-\theta)^{2\sum x_{1,i}} (1-\theta^2)^{\sum x_{2,i}} (1+\theta)^{2\sum x_{3,i}}}{2^{2\sum x_{1,i}} 2^{\sum x_{2,i}} 2^{2\sum x_{3,i}}} = \frac{(1-\theta)^{2Y_1} (1-\theta^2)^{Y_2} (1+\theta)^{2Y_3}}{2^{2Y_1} 2^{Y_2} 2^{2Y_3}}$$

$$\ln(L(\theta|x)) = 2Y_1 \ln(1-\theta) + Y_2 \ln(1-\theta^2) + 2Y_3 \ln(1+\theta) - (2Y_1 + Y_2 + 2Y_3) \ln(2)$$

$$\frac{d}{d\theta} \ln(L(\theta|x)) = -\frac{2Y_1}{1-\theta} - \frac{2Y_2\theta}{1-\theta^2} + \frac{2Y_3}{1+\theta} = 0 = -\frac{Y_1}{1-\theta} - \frac{Y_2\theta}{1-\theta^2} + \frac{Y_3}{1+\theta} \implies -\frac{Y_1(1-\theta^2)(1+\theta) - Y_2\theta(1-\theta)(1+\theta) + Y_3(1-\theta)(1-\theta^2)}{STUFF} = 0$$

$$\implies -Y_1(1-\theta^2 + \theta - \theta^3) - Y_2(\theta - \theta^3) + Y_3(1-\theta - \theta^2 + \theta^3) = 0$$

$$\implies (Y_3 - Y_1) - \theta(Y_1 + Y_2 + Y_3) + \theta^2(Y_1 - Y_3) + \theta^3(Y_1 + Y_2 + Y_3) = 0$$

$$\text{Since } \sum_{j=1}^3 = m = 1 \implies Y_1 + Y_2 + Y_3 = n$$

$$\implies (Y_3 - Y_1) - n\theta - \theta^2(Y_3 - Y_1) + n\theta^3 = (Y_3 - Y_1)(1 - \theta^2) + n\theta(\theta^2 - 1) = (Y_1 - Y_3)(\theta^2 - 1) + n\theta(\theta^2 - 1) = 0$$

$$\implies (Y_1 - Y_3) + n\theta = 0 \implies \hat{\theta} = \frac{Y_3 - Y_1}{n}$$

b)

Find, as simple form as possible, asymptotic size α rejection regions for Generalized LRT for $H_0 : \theta = 0, H_1 : \theta \neq 1$

We just found the unrestricted MLE. To find the restricted MLE, we are saying that $\hat{\theta}_R = 0$, so we will just plug that in anywhere we have 0.

$$\text{LRT} = \frac{L(\hat{\theta}_R|x)}{L(\hat{\theta}|x)} = \frac{\frac{(1-0)^{2Y_1} (1-0^2)^{Y_2} (1+0)^{2Y_3}}{2^{2Y_1+Y_2+2Y_3}}}{\frac{(1-\hat{\theta})^{2Y_1} (1-\hat{\theta}^2)^{Y_2} (1+\hat{\theta})^{2Y_3}}{2^{2Y_1+Y_2+2Y_3}}} = \frac{1}{(1-\hat{\theta})^{2Y_1} (1-\hat{\theta}^2)^{Y_2} (1+\hat{\theta})^{2Y_3}}$$

$$\text{Thus, we reject anytime } \frac{1}{(1-\hat{\theta})^{2Y_1} (1-\hat{\theta}^2)^{Y_2} (1+\hat{\theta})^{2Y_3}} \notin (c_1, c_2).$$

We can then use the Wilks' Theorem: $-2\ln(\lambda) \xrightarrow{D} \chi_1^2$

So our region is rejecting when $\lambda \notin (k_1, k_2)$ where k_1 fulfills: $P(\chi_1^2 \leq -2\ln(k_1)) = \alpha_1$ and k_2 fulfills: $P(\chi_1^2 \geq -2\ln(k_2)) = \alpha_2$, and $\alpha_1 + \alpha_2 = \alpha$

c)

```
n <- 100
theta <- seq(-9, 9, by = 1)/10
alpha <- .05
# Assuming equal sided
lower <- qchisq(alpha/2, 1)
upper <- qchisq(1-alpha/2, 1)
n_sim <- 500

power <- numeric(length(theta))
for (i in 1:length(theta)) {
  p1 <- (1-theta[[i]])^2/4
```

```

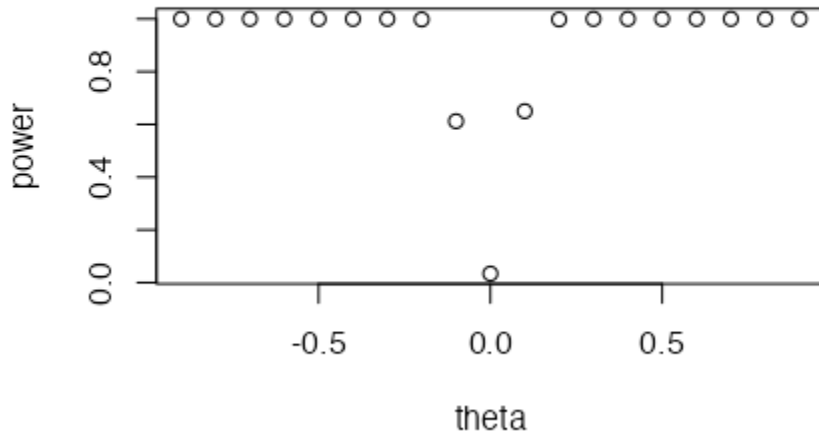
p2 <- (1-theta[[i]]^2)/2
p3 <- (1+theta[[i]]^2)/4

val <- numeric(length(n_sim))
for (j in 1:n_sim) {
  dt <- rmultinom(n, size = 3, prob = c(p1,p2,p3))
  Y1 <- sum(dt[1,])
  Y2 <- sum(dt[2,])
  Y3 <- sum(dt[3,])
  theta_hat <- (Y3 - Y1)/n
  val[[j]] <- (1-theta_hat)^(2*Y1)*(1-theta_hat^2)^(Y2)*(1+theta_hat)^(2*Y3)
}

power[[i]] <- mean(val > upper | val < lower)
}

plot(theta, power)

```



We can see that as θ moves away from the Null hypothesis, the Power increases and approaches 1.